Kinetics: linear motion

- Newton's second law
    - Conservation of linear momentum
- Newton's third law

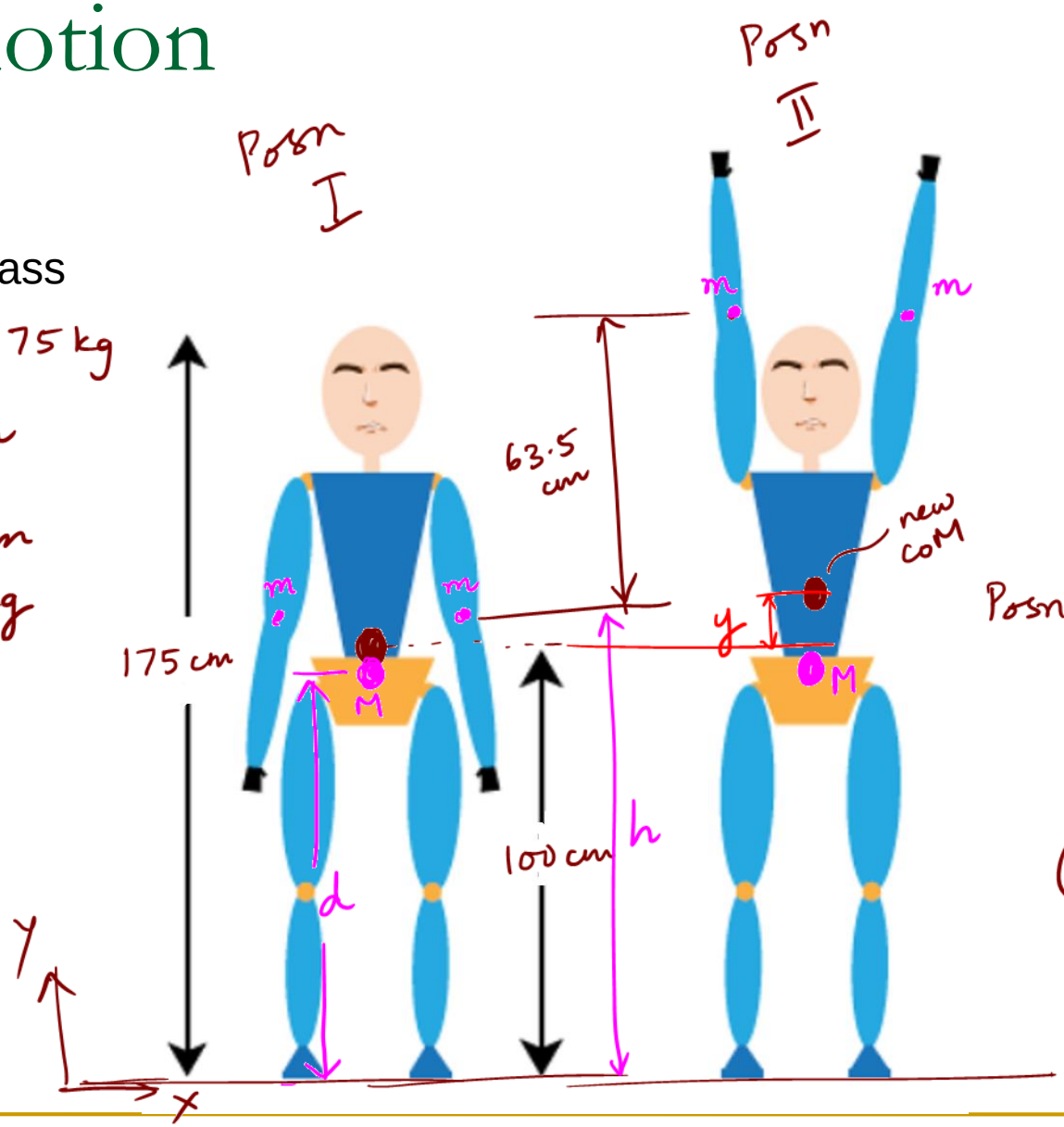
Linear motion

- Centre of mass

Mass of person = 75 kg

H = 175 cm

mass of each arm = 3 kg



Mass of each arm = m

Mass of rest of the body = M

CoM of arms moves upward by 63.5 cm when the person raises the arms

Find the new CoM for the whole body

Posn I : $(M+2m)100 = Md + 2mh$

$(M+2m)(100+y) = Md + 2m(h+63.5)$

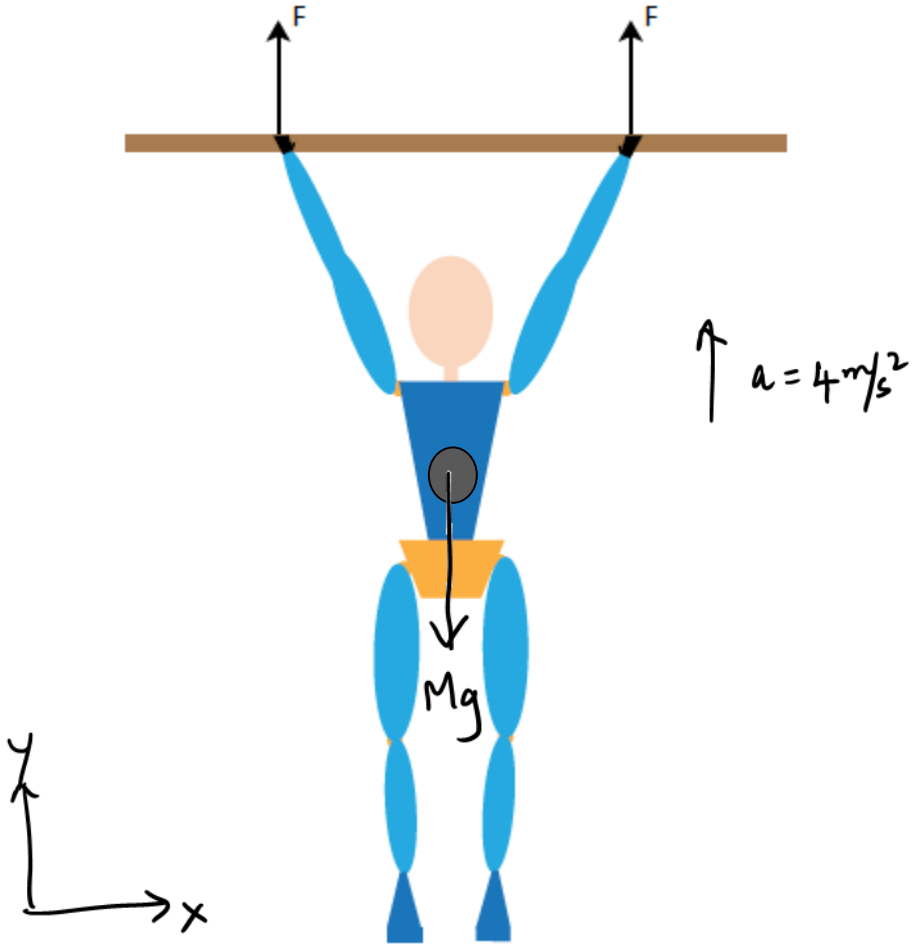
$(M+2m)y = 2m \times 63.5 = 127m$

$y = 5.08$ cm.

Linear motion

Person doing chin-ups / pull-ups

75 kg working out the biceps & lattisimus muscles



If during the rising phase, the person's acc^{n}. reaches a peak of 4 m/s^2, determine the contact forces exerted by the bar on the athlete

$$\sum F_x = m a_x = 0$$

$$\sum F_y = m a_y$$

$$2F - Mg = M a_y$$

$$\therefore \quad 2F = M(g + a_y) = 75(10 + 4)$$

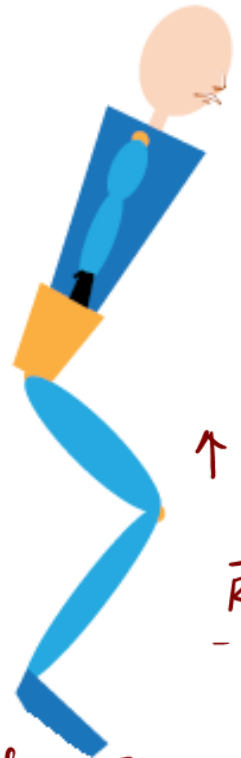
$$\therefore \quad F = 517 \text{ N}$$

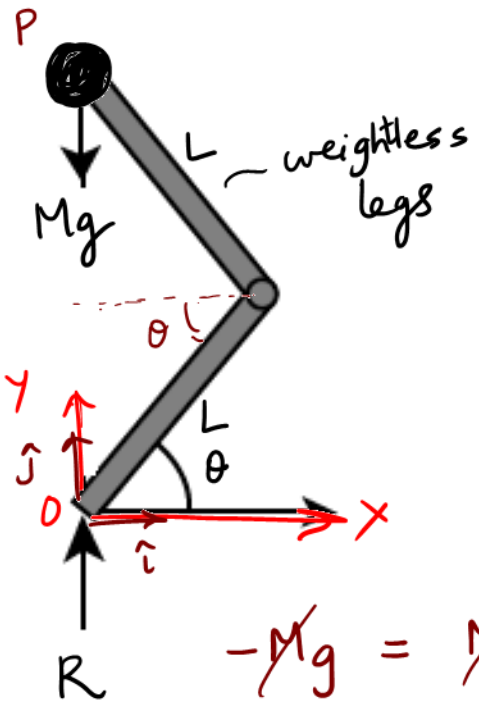
Static case $F = \frac{75g}{2} = 368$ N

Linear motion

Model : Mass M attached to weightless legs

$L = 0.5\,m$





$\bar{r}_{P/O} = 2L\sin\theta\,\hat{j}$ ——— (1)

$\bar{V}_{P/O} = \frac{d\bar{r}_{P/O}}{dt} = 2L\frac{d}{dt}\sin\theta\,\hat{j}$

$= 2L\frac{d}{d\theta}(\sin\theta)\cdot\frac{d\theta}{dt}\,\hat{j}$

$= 2L\cos\theta\cdot\frac{d\theta}{dt}\,\hat{j}$ ——— (2)

$\bar{a} = \frac{d\bar{V}_{P/O}}{dt} = 2L(-\sin\theta)\left(\frac{d\theta}{dt}\right)^2\hat{j} + 2L\cos\theta\,\frac{d^2\theta}{dt^2}\,\hat{j}$

$\uparrow \quad R - Mg = M a_y$

$\bar{R} + M\bar{g} = M\bar{a}$

At the instant the jumper leaves the ground, $\bar{R} = 0$, $\theta = 60^\circ$, $\frac{d^2\theta}{dt^2} = 0$

$-Mg = M\left(-2L\sin\theta\left(\frac{d\theta}{dt}\right)^2 + 2L\cos\theta\,\frac{d^2\theta}{dt^2}\right)$

$\bar{V}$ at take-off $=$ $\uparrow$

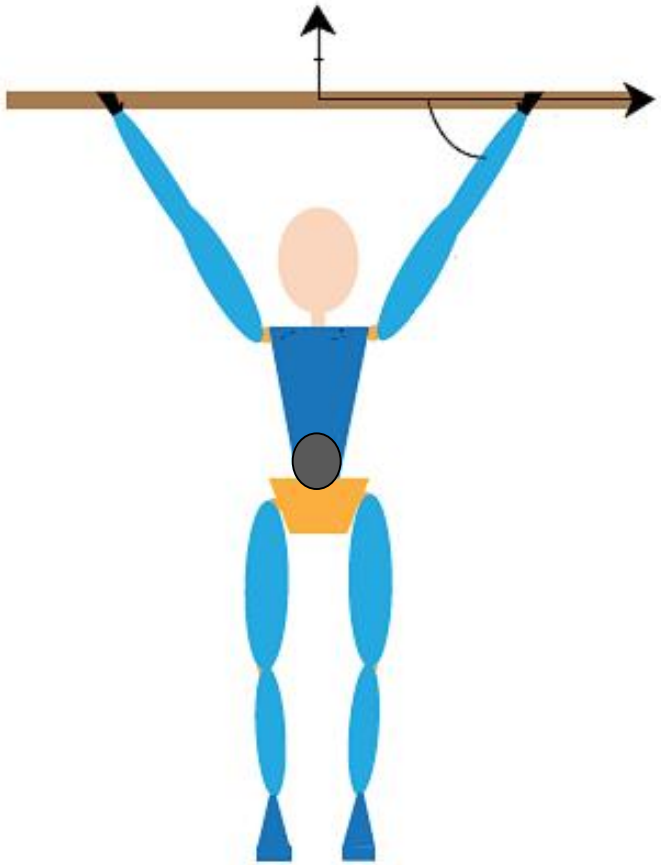
$acc \uparrow = -g$

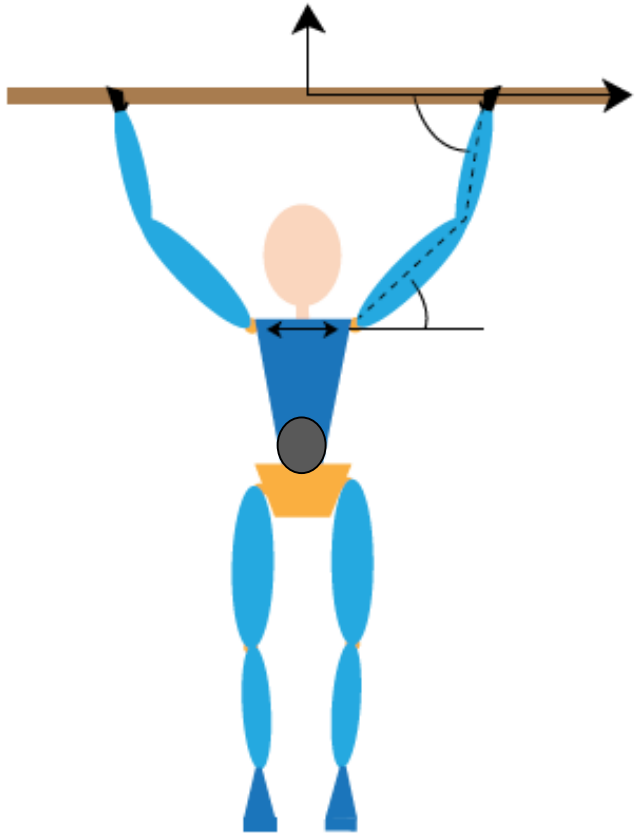
$s = \frac{V^2}{2g}$

$\theta = 60^\circ, \frac{d\theta}{dt} = 3.3$ rad/s

for $\theta = 90^\circ$, $\frac{d\theta}{dt} = 3.13$ rad/s

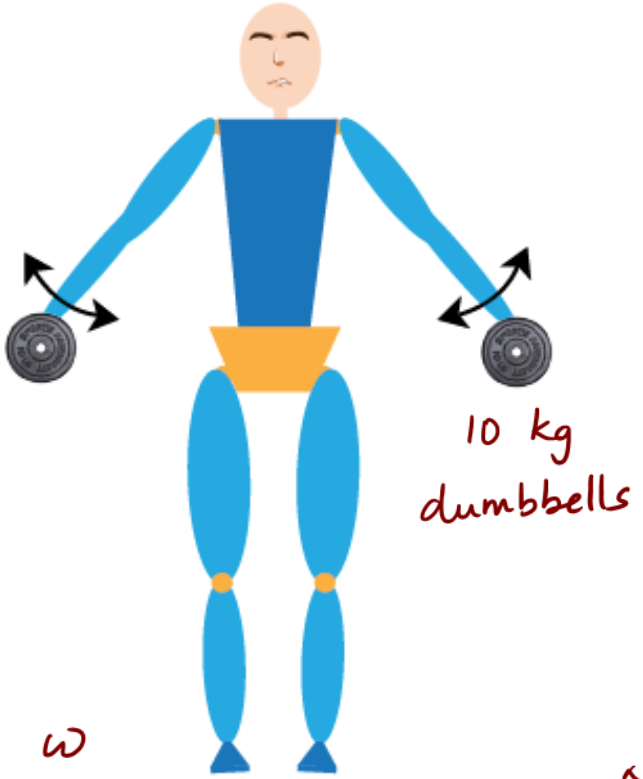
Linear motion



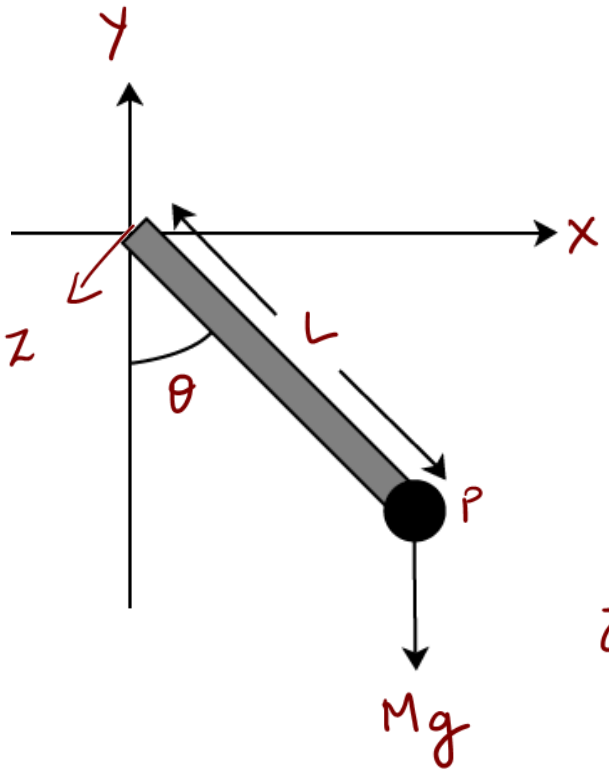


Angular motion

Weightless rod to which mass M is attached



10 kg dumbbells



$$\bar{r}_{P/O} = L\sin\theta\,\hat{i} - L\cos\theta\,\hat{j}$$

$$\bar{v}_{P/O} = \frac{d\bar{r}_{P/O}}{dt} = L\cos\theta\,\frac{d\theta}{dt}\,\hat{i} + L\sin\theta\,\frac{d\theta}{dt}\,\hat{j}$$

$$= \frac{d\theta}{dt}\,\hat{k} \times \bar{r}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \frac{d\theta}{dt} \\ LS_\theta & -LC_\theta & 0 \end{vmatrix} = \hat{i}\left(+LC_\theta\frac{d\theta}{dt}\right) - \hat{j}\left(-LS_\theta\frac{d\theta}{dt}\right) + \hat{k}\,0$$

$$= \bar{v}_{P/O}$$

denote

$\frac{d\theta}{dt} = \omega$

$\frac{d^2\theta}{dt^2} = \alpha$

$\frac{d\theta}{dt}\,\hat{k}$

$$\bar{a}_{P/O} = \frac{d\bar{v}_{P/O}}{dt}$$

$$= \frac{d^2\theta}{dt^2}\,\hat{k}\times\bar{r} + \frac{d\theta}{dt}\,\hat{k}\times\frac{d\bar{r}}{dt}$$

$$= \frac{d^2\theta}{dt^2}\,\hat{k}\times\bar{r} + \frac{d\theta}{dt}\,\hat{k}\times\left(\frac{d\theta}{dt}\,\hat{k}\times\bar{r}\right)$$

TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2